

Predictive Modelling

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2: Basic concepts, a start

Outline

Autocorrelation

White noise

Conditional and unconditional distribution

Persistence of shocks

Autoregression of order 1 (to begin with)

Mean and intercept

A key feature of time series

Key property of time series: sequence of observations is determined by calendar time. That is, y_t always comes after y_{t-1} .

Hence, there is a natural ordering of the observations.

The values of the variable y at consecutive points in time are often closely related. That is, there often is a substantial correlation between y_t and y_{t-1}

There often is non-zero correlation between y_t and y_{t-k} , also for $k > 1$.

The correlation between y_t and lagged values of the variable itself, y_{t-k} , is called autocorrelation.

Computing autocorrelations from actual data

First order autocorrelation (note the start at $t = 2$):

$$\hat{\rho}_1 = \frac{\frac{1}{T-1} \sum_{t=2}^T (y_t - \bar{y}_t)(y_{t-1} - \bar{y}_{t-1})}{\sqrt{\frac{1}{T-1} \sum_{t=2}^T (y_t - \bar{y}_t)^2} \sqrt{\frac{1}{T-1} \sum_{t=2}^T (y_{t-1} - \bar{y}_{t-1})^2}}$$

Note that in practice:

$$\bar{y}_t \approx \bar{y}_{t-1}$$

$$\frac{1}{T-1} \sum_{t=2}^T (y_t - \bar{y}_t)^2 \approx \frac{1}{T-1} \sum_{t=2}^T (y_{t-1} - \bar{y}_{t-1})^2$$

So, usually we compute

$$\hat{\rho}_1 = \frac{\sum_{t=2}^T (y_t - \bar{y})(y_{t-1} - \bar{y})}{\sum_{t=1}^T (y_t - \bar{y})^2}$$

with

$$\bar{y} = \frac{1}{T} \sum_{t=1}^T y_t$$

In general,

The k -th autocorrelation

$$\hat{\rho}_k = \frac{\hat{\gamma}_k}{\hat{\gamma}_0}$$

with the k -th autocovariance

$$\hat{\gamma}_k = \frac{1}{T} \sum_{t=k+1}^T (y_t - \bar{y})(y_{t-k} - \bar{y})$$

γ_0 is the variance.

Often, one computes the Empirical Autocorrelation Function for $k = 1, 2, \dots, K$, with for example $K = 10$ or $K = 30$.

Statistically significant autocorrelations

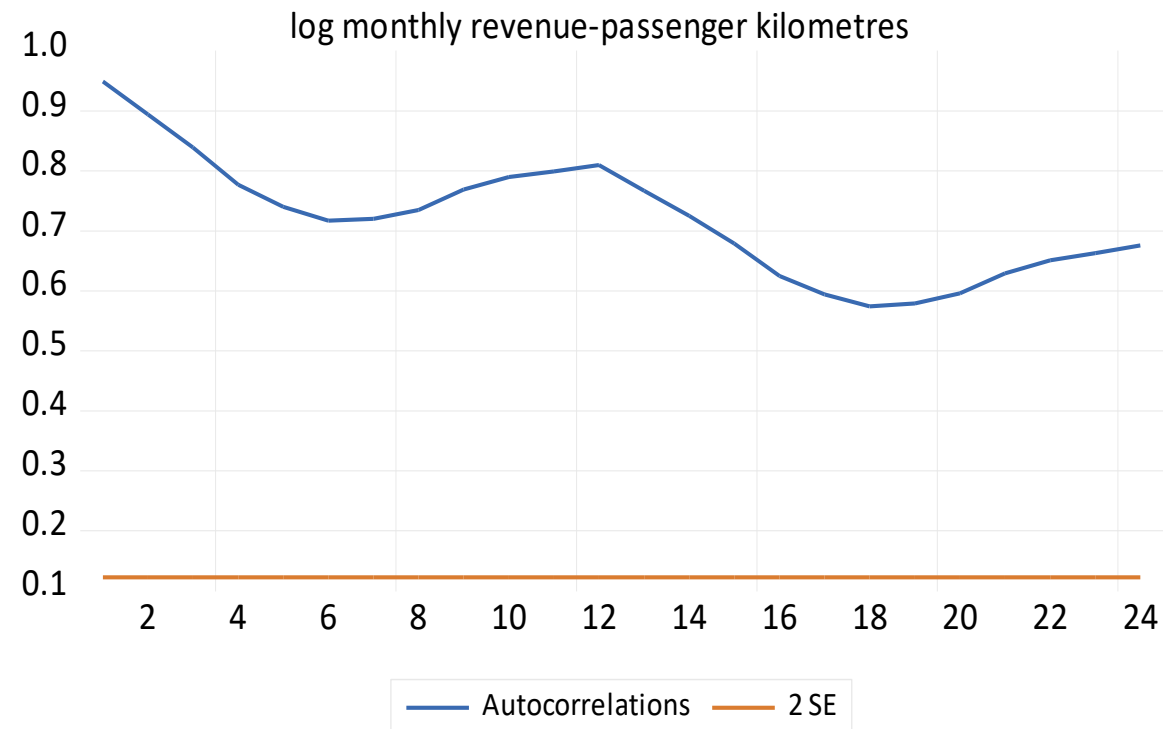
The standard error of autocorrelations is

$$\frac{1}{\sqrt{T}}$$

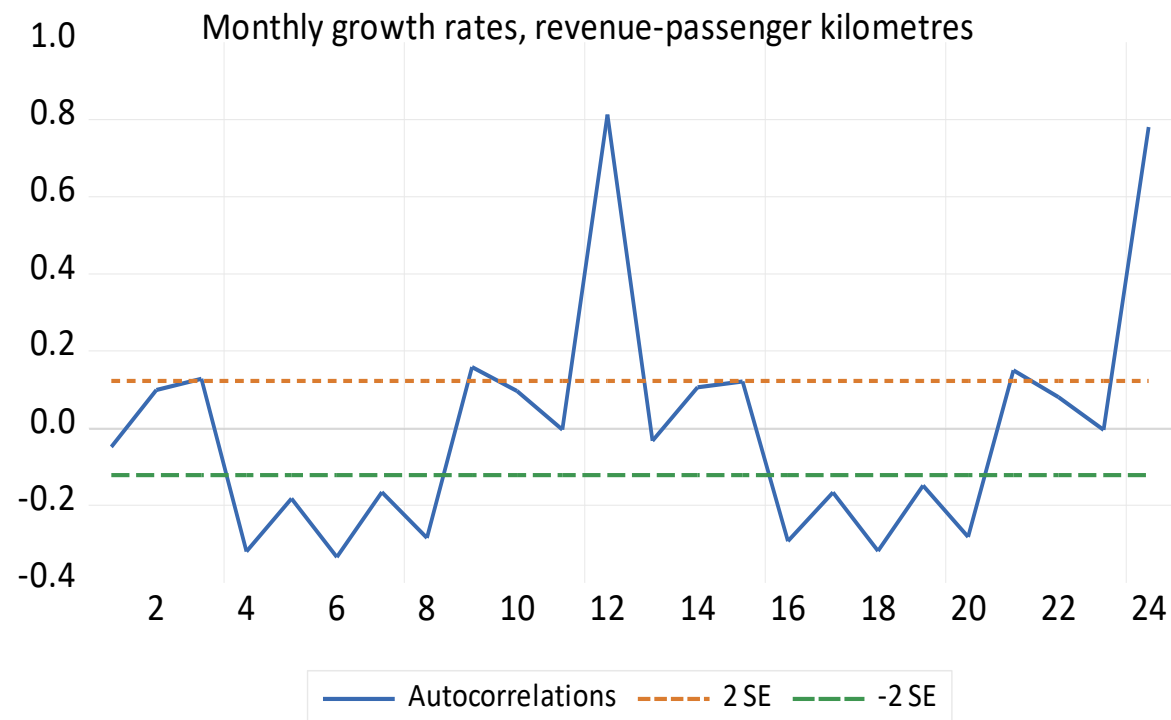
Taking a 5% confidence level, then (approximately) ρ_k is significant when

$$\hat{\rho}_k > \frac{2}{\sqrt{T}} \text{ or } \hat{\rho}_k < -\frac{2}{\sqrt{T}}$$

Example, log levels, autocorrelations



Example, monthly growth rates



Time series models

Empirical autocorrelation functions can take many different shapes.

Important!: A time series model implies certain autocorrelation properties for the time series that is modelled.

The empirical autocorrelations can help to find $g(\cdot)$ and any useful y_{t-k} in the function

$$g(y_{t-1}, y_{t-2}, \dots, y_{t-p}; \theta)$$

Shocks (news)

The shocks (news) ε_t are very important.

They are responsible for the fact that, even if we know the true functional form $g(\cdot)$ as well as the true parameter values, we cannot perfectly predict the future value y_{T+1} based upon the past and present observations $y_T, y_{T-1}, \dots$

ε_t is responsible for the random behavior of y_t .

How a time series responds to such shocks is related to the autocorrelations.

Assumptions about ε_t

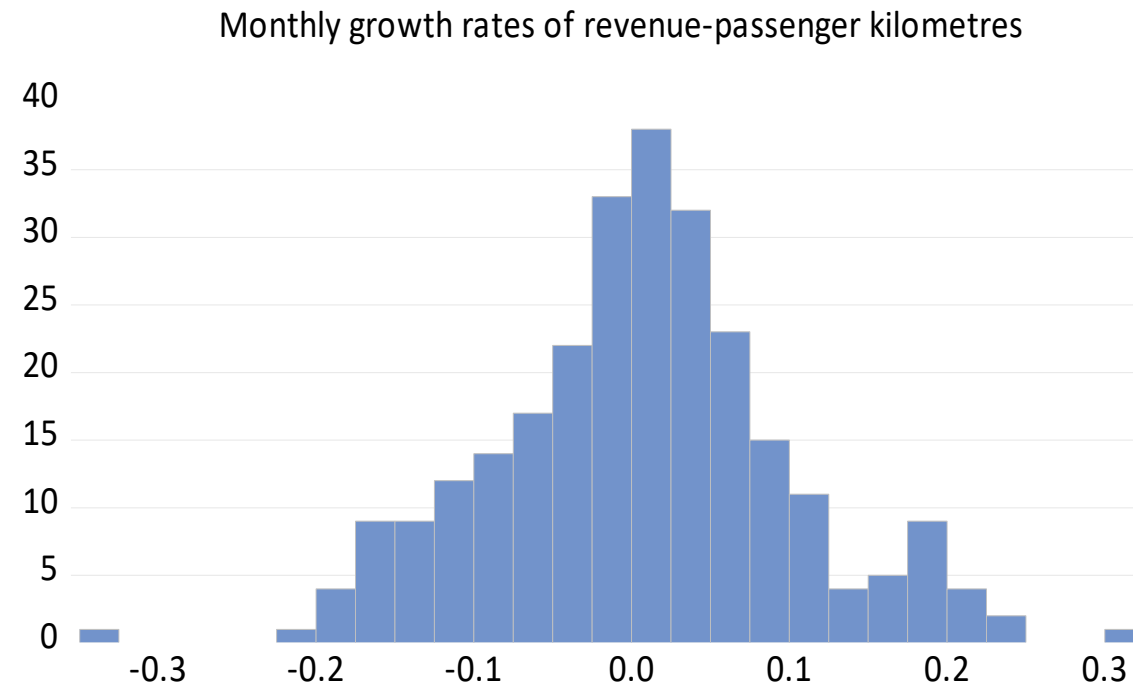
A time series ε_t is called white noise if for all $t = 1, 2, \dots, T$

$$\begin{aligned}E(\varepsilon_t) &= 0 \\E(\varepsilon_t^2) &= \sigma^2 \\E(\varepsilon_s \varepsilon_t) &= 0, s \neq t\end{aligned}$$

We are satisfied with $g(\cdot)$ if the estimated shocks have these three white noise properties.

Unconditional distribution

Unconditional distribution is also known as marginal distribution, and it reflects the probabilities all occurrences ignoring any knowledge gained from previous or external events. The unconditional distribution is constant over time.



Conditional distribution

The conditional distribution of y_t is the distribution given the information set at time $t - 1$. This is useful for forecasting.

Given

$$y_t = g(y_{t-1}, y_{t-2}, \dots, y_{t-p}; \theta) + \varepsilon_t$$

the information set consists of $g(y_{t-1}, y_{t-2}, \dots, y_{t-p}; \theta)$, and hence

Expected value of y_t given $g(y_{t-1}, y_{t-2}, \dots, y_{t-p}; \theta) = g(y_{t-1}, y_{t-2}, \dots, y_{t-p}; \theta)$

Variance of y_t given $g(y_{t-1}, y_{t-2}, \dots, y_{t-p}; \theta) = \sigma^2$

An autoregression of order 1

A first order autoregressive model is given by

$$y_t = \phi_1 y_{t-1} + \varepsilon_t$$

This implies that (with some starting value y_0)

$$y_1 = \phi_1 y_0 + \varepsilon_1$$

$$y_2 = \phi_1 y_1 + \varepsilon_2 = \phi_1 (\phi_1 y_0 + \varepsilon_1) + \varepsilon_2 = \phi_1^2 y_0 + \phi_1 \varepsilon_1 + \varepsilon_2$$

$$y_3 = \phi_1 y_2 + \varepsilon_3 = \phi_1 (\phi_1^2 y_0 + \phi_1 \varepsilon_1 + \varepsilon_2) + \varepsilon_3$$

...

$$y_t = \phi_1^t y_0 + \phi_1^{t-1} \varepsilon_1 + \phi_1^{t-2} \varepsilon_2 + \cdots + \varepsilon_t$$

An AR(1), continued

$$y_t = \phi_1^t y_0 + \phi_1^{t-1} \varepsilon_1 + \phi_1^{t-2} \varepsilon_2 + \cdots + \varepsilon_t$$

In short-hand:

$$y_t = \phi_1^t y_0 + \sum_{i=0}^{t-1} \phi_1^i \varepsilon_{t-i}$$

The effect of shock ε_{t-i} on y_t is determined by ϕ_1^i

When $|\phi_1| < 1$, ϕ_1^i approaches zero with increasing i .

Effects of shocks

$$y_t = \phi_1^t y_0 + \sum_{i=0}^{t-1} \phi_1^i \varepsilon_{t-i}$$

Transitory if $|\phi_1| < 1$, as then ϕ_1^i approaches zero with increasing i .

Explosive if $|\phi_1| > 1$, as then ϕ_1^i gets larger with increasing i .

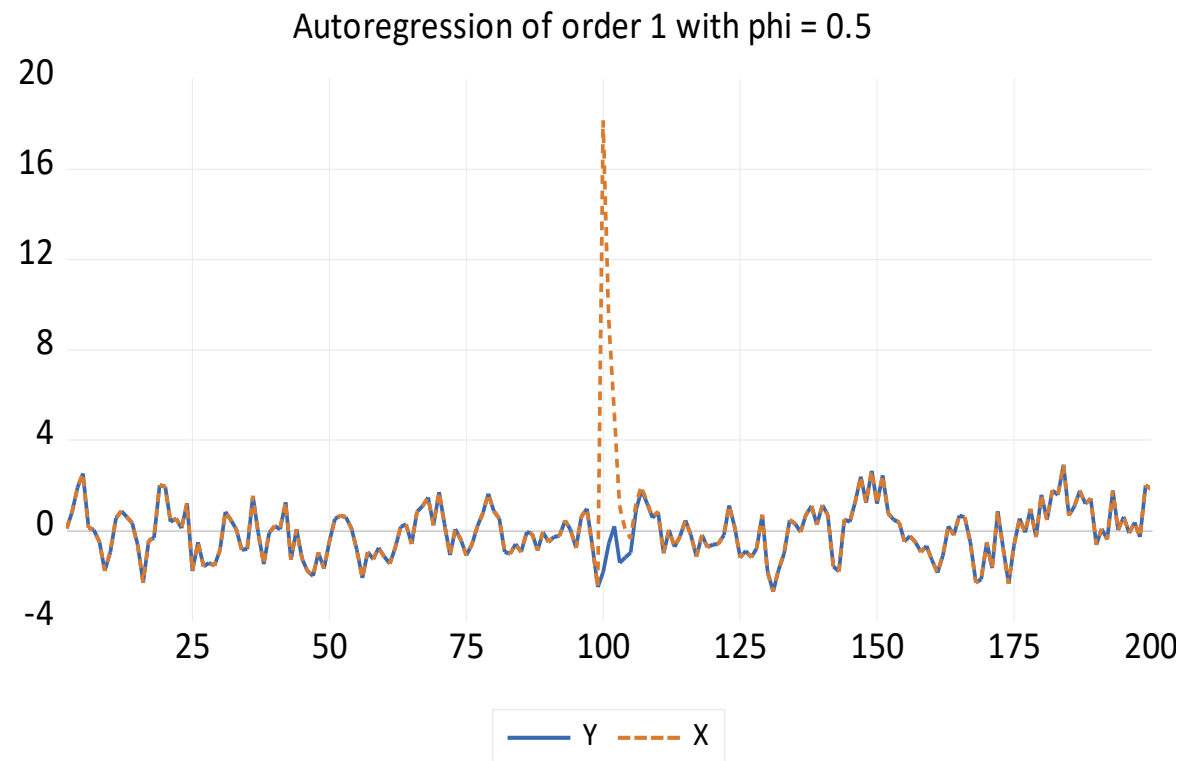
Permanent if $|\phi_1| = 1$, because then (and this is a special case to be discussed more in detail later)

$$y_t = y_0 + \sum_{i=0}^{t-1} \varepsilon_{t-i}$$

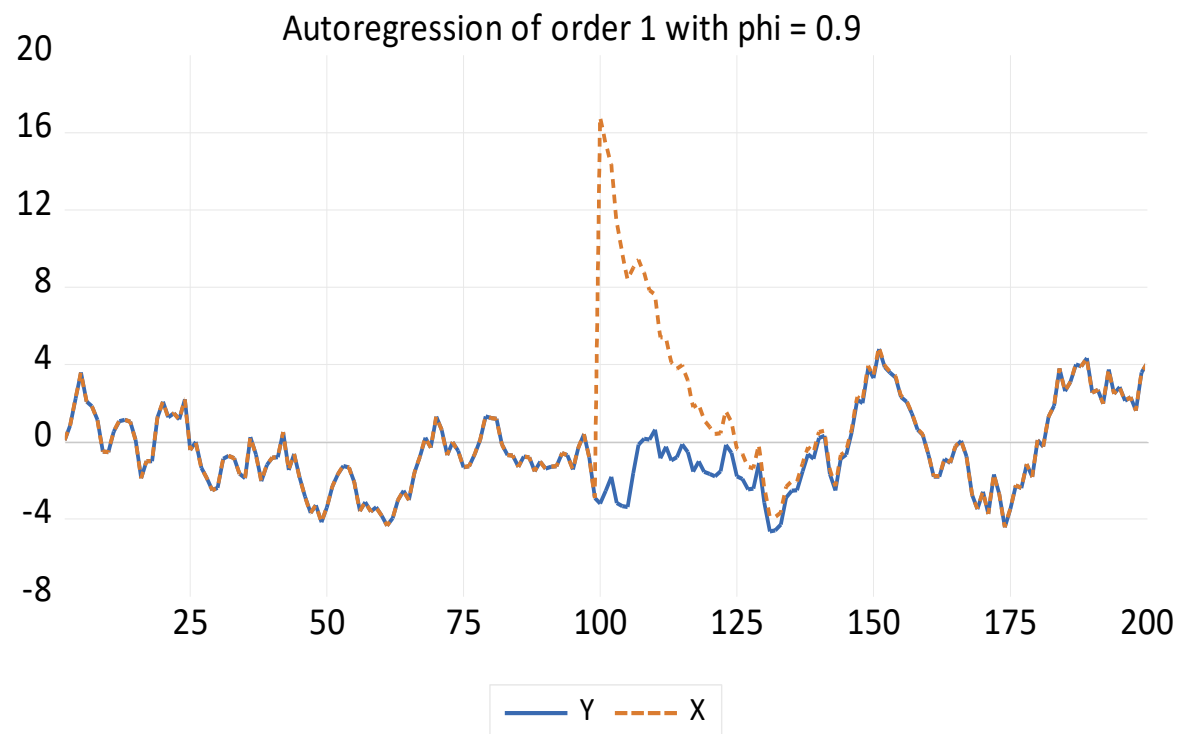
Exercise 1 (using R)

This exercise shows that the impact of shocks depends on the value of the AR(1) parameter ϕ_1 .

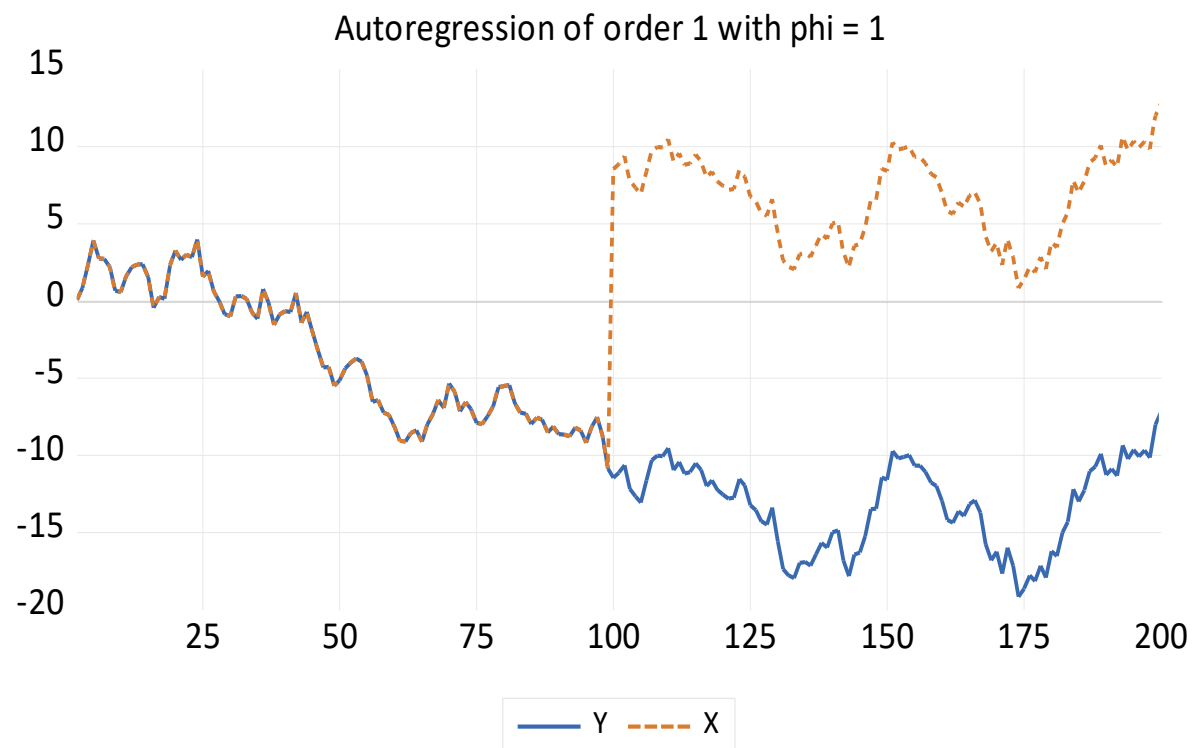
Effect of large shock, $\phi_1 = 0.5$



Effect of large shock, $\phi_1 = 0.9$



Effect of large shock, $\phi_1 = 1.0$



Exercise 2 (using R)

How does the empirical autocorrelation function look like for an autoregression of order 1?

Recall:

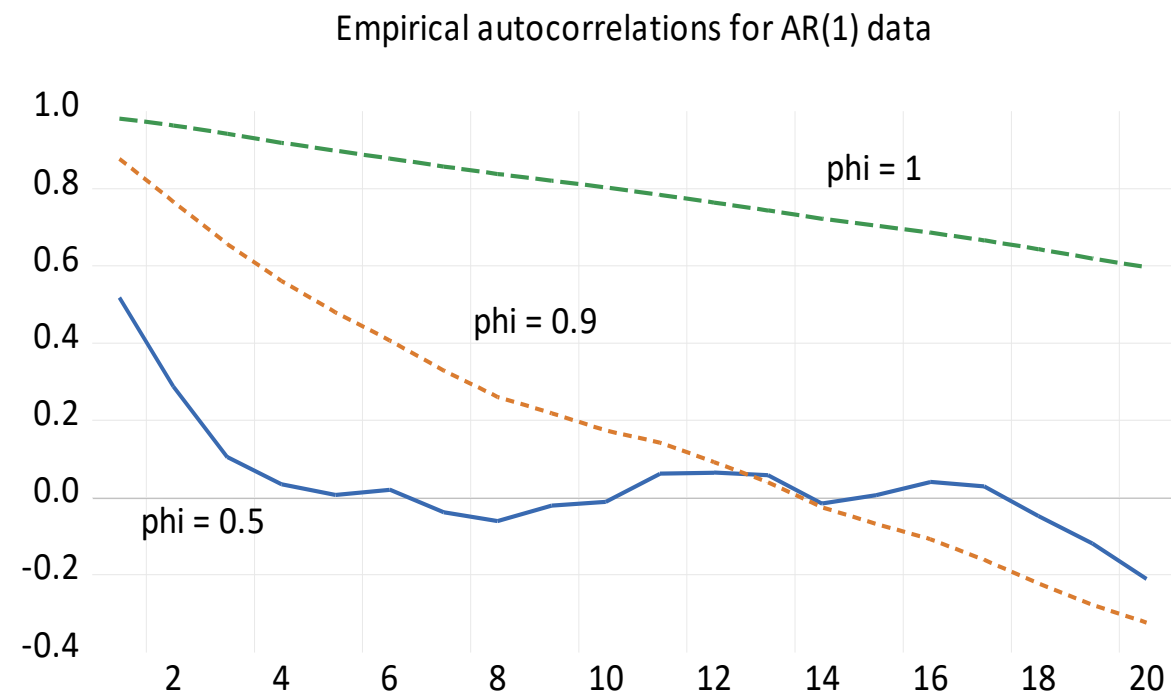
The k -th autocorrelation

$$\hat{\rho}_k = \frac{\hat{\gamma}_k}{\hat{\gamma}_0}$$

with the k -th autocovariance

$$\hat{\gamma}_k = \frac{1}{T} \sum_{t=k+1}^T (y_t - \bar{y})(y_{t-k} - \bar{y})$$

Empirical autocorrelations



Conditional and unconditional mean for an AR(1)

Start with

$$y_t = \phi_1 y_{t-1} + \varepsilon_t$$

with $|\phi_1| < 1$ and (!) $E(y_t) = E(y_{t-1})$

Conditional mean:

$$\phi_1 y_{t-1}$$

Unconditional mean:

$$\begin{aligned} E(y_t) &= \phi_1 E(y_{t-1}) + E(\varepsilon_t) \\ E(y_t) &= \phi_1 E(y_t) + 0 \\ (1 - \phi_1)E(y_t) &= 0 \\ E(y_t) &= 0 \end{aligned}$$

Now include an intercept

$$y_t = \delta + \phi_1 y_{t-1} + \varepsilon_t$$

Then, again with $|\phi_1| < 1$ and (!) $E(y_t) = E(y_{t-1})$

$$E(y_t) = \delta + \phi_1 E(y_{t-1}) + E(\varepsilon_t)$$

$$(1 - \phi_1)E(y_t) = \delta + 0$$

$$E(y_t) = \frac{\delta}{1 - \phi_1}$$

Other way of writing

With unconditional mean μ

$$\mu = E(y_t) = \frac{\delta}{1 - \phi_1}$$

and hence

$$\delta = \mu(1 - \phi_1)$$

then

$$y_t = \delta + \phi_1 y_{t-1} + \varepsilon_t$$

can be written as

$$y_t = \mu(1 - \phi_1) + \phi_1 y_{t-1} + \varepsilon_t$$

and after rearranging as

$$y_t - \mu = \phi_1(y_{t-1} - \mu) + \varepsilon_t$$

Conditional and unconditional mean

Conditional mean is

$$\mu(1 - \phi_1) + \phi_1 y_{t-1}$$

or

$$\delta + \phi_1 y_{t-1}$$

Unconditional mean is

$$\mu$$

Mean reversion

$$y_t - \mu = \phi_1(y_{t-1} - \mu) + \varepsilon_t$$

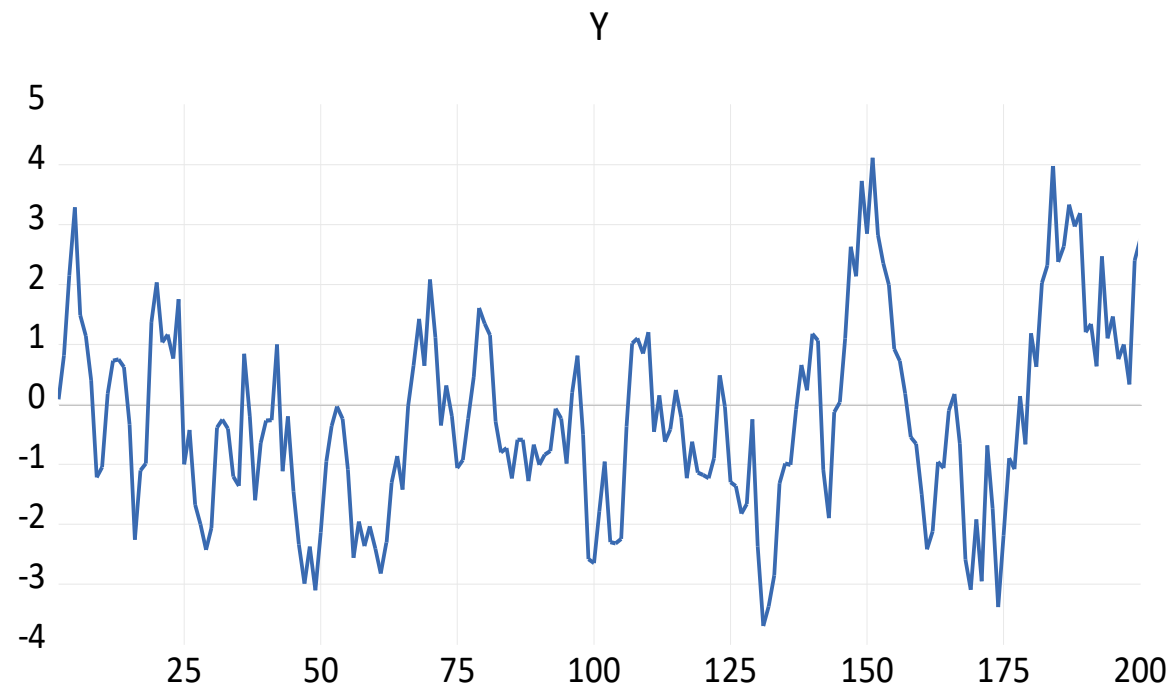
With $0 < \phi_1 < 1$, if

$$y_{t-1} > \mu \quad \text{or} \quad y_{t-1} < \mu$$

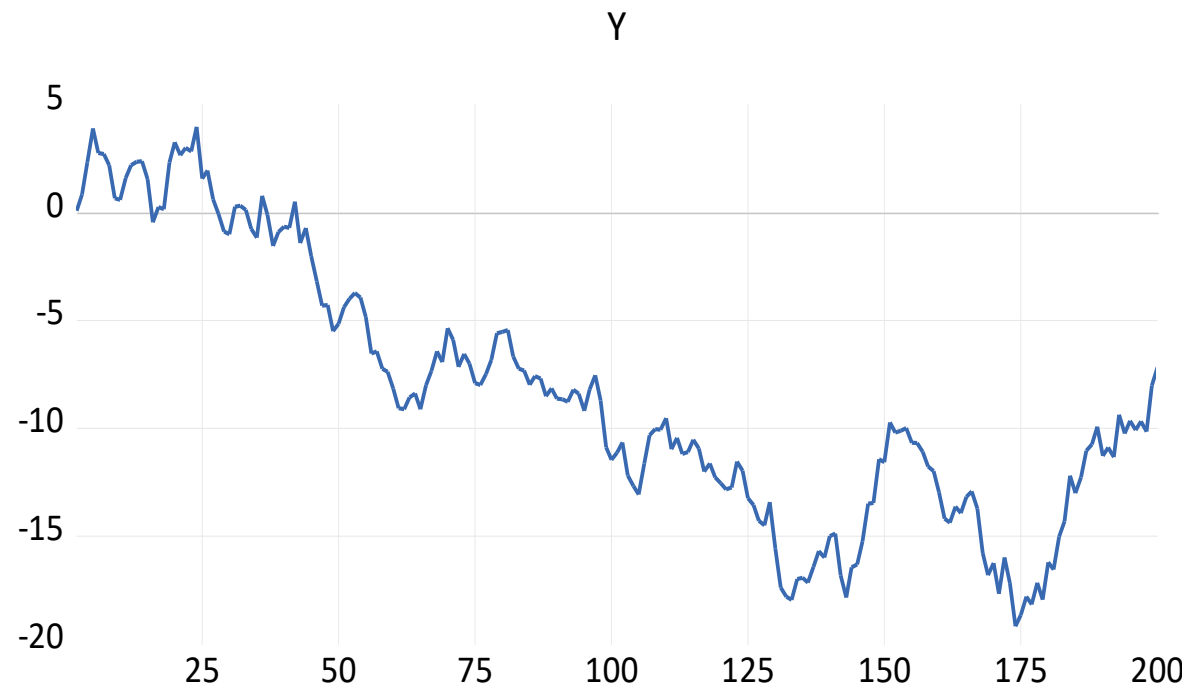
then

$$|y_t - \mu| < |y_{t-1} - \mu|$$

Mean reversion: $\phi_1 = 0.8$



No mean reversion: $\phi_1 = 1.0$



Theoretical autocorrelation function, AR(1)

The variance is

$$\gamma_0 = E(y_t - \mu)(y_t - \mu)$$

To save notation, set $\mu = 0$, then

$$\gamma_0 = E(y_t^2) = E(\phi_1 y_{t-1})(\phi_1 y_{t-1}) + E(\varepsilon_t^2) + 2E(\phi_1 y_{t-1} \varepsilon_t)$$

With $\gamma_0 = E(y_t^2) = E(y_{t-1}^2)$ and $E(y_{t-1} \varepsilon_t) = 0$, we get

$$E(y_t^2) = \phi_1^2 E(y_{t-1}^2) + \sigma^2 + 0$$

And hence

$$\gamma_0 = \frac{\sigma^2}{1 - \phi_1^2}$$

The closer ϕ_1 is to 1, the larger the variance.

First order autocorrelation

First order autocovariance

$$\gamma_1 = E(y_t y_{t-1}) = E(\phi_1 y_{t-1} + \varepsilon_t) y_{t-1} = \phi_1 E(y_{t-1}^2) + 0 = \phi_1 \gamma_0$$

First order autocorrelation is

$$\rho_1 = \frac{\gamma_1}{\gamma_0} = \frac{\phi_1 \gamma_0}{\gamma_0} = \phi_1$$

More autocorrelations

k -th order autocovariance

$$\gamma_k = E(y_t y_{t-k}) = E(\phi_1 y_{t-1} + \varepsilon_t) y_{t-k} = \phi_1 \gamma_{k-1}$$

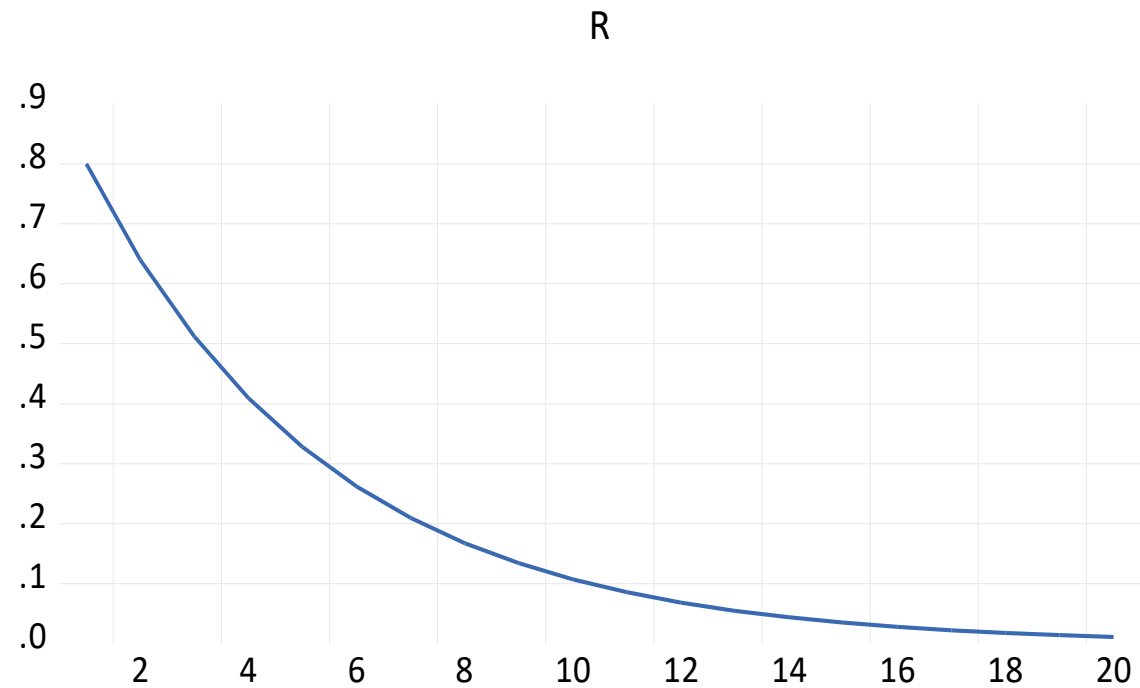
k -th order autocorrelation

$$\rho_k = \frac{\gamma_k}{\gamma_0} = \frac{\gamma_{k-1}}{\gamma_0} = \phi_1 \rho_{k-1}$$

Hence,

$$\begin{aligned}\rho_2 &= \phi_1 \rho_1 = \phi_1^2 \\ \rho_3 &= \phi_1^3\end{aligned}$$

Theoretical pattern, ACF of AR(1) with $\phi_1 = 0.8$



Parameter estimation

Apply Ordinary Least Squares to

$$y_t = \delta + \phi_1 y_{t-1} + \varepsilon_t$$

For this it holds that

$$E(\hat{\phi}_1) \approx \phi_1 - \frac{1 + 3\phi_1}{T - 1}$$

So, there is a small sample bias towards 0

What did we learn?

An AR(1) implies a specific pattern for autocorrelations (note: for other models, other patterns will appear)

Empirical autocorrelations can be computed and matched with the theoretical autocorrelations to “identify” a model (the function $g(\cdot)$)

Shocks drive a time series. Shocks cannot be predicted. The impact of shocks depends on the parameters in the model.